



Impact of Social Media Usage on Productivity and Stress

A Statistical Project Using Regression, Hypothesis Testing, and Clustering

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1 Introduction

Social media platforms have become an important part of everyday life, affecting how individuals communicate, work, and manage their time. Although these platforms offer advantages such as rapid communication and easy access to information, they also introduce frequent interruptions through notifications and continuous digital engagement. Such disruption may affect concentration levels, contribute to psychological strain, and influence individual productivity.

The main research question of this study is:

How does social media usage relate to individual productivity and stress levels?

This research question is relevant because productivity and well-being are important factors in both academic and professional environments. As daily social media usage continues to increase, understanding how this behavior is associated with productivity and stress becomes increasingly important for students, employees, and organizations.

Previous research has examined the relationship between social media use and work-related outcomes. Studies have shown that personal social media usage during work hours is associated with reduced efficiency and lower well-being [1]. Other research indicates that frequent digital interruptions and online distractions are linked to higher stress levels and reduced psychological well-being [2]. From a cognitive perspective, repeated task switching caused by smartphone notifications may interfere with sustained attention and self-regulation processes [3].

In this project, a large behavioral dataset is analyzed to explore the relationship between social media usage, productivity, and stress. Exploratory data analysis, correlation analysis, hypothesis testing, and regression models are used to examine these relationships and assess their statistical properties.

2 Data Source

The dataset used in this study is the *Social Media vs Productivity* dataset made publicly available on Kaggle [?]. The dataset contains 30,000 observations and simulates realistic behavioral patterns related to social media usage, work habits, and individual well-being. The dataset includes demographic information such as age and gender, along with behavioral and lifestyle variables. The main variables used in this analysis include daily social media usage time, productivity scores, stress levels, sleep duration, coffee consumption, and the number of burnout days per month. Productivity is measured on a scale from 0 to 10 and represents a simulated performance score.

The dataset is appropriate for this analysis because it combines indicators of digital

behavior with measures of productivity and stress. In addition, the presence of missing values and substantial variability across observations makes it suitable for demonstrating practical data preprocessing and statistical analysis techniques.

It is important to note that the dataset is simulated rather than collected from real individuals, which limits the generalizability of the findings. Furthermore, the data are cross-sectional, meaning that observed relationships cannot be interpreted as causal.

3 Data Preprocessing

Before conducting statistical analyses, the dataset was prepared using the following steps:

1. **Missing value handling** — The dataset was inspected for missing values in key analytical variables, including daily social media usage, productivity score, stress level, sleep hours, coffee intake, and burnout days. Records containing missing values in any of these variables were removed to avoid biased results in correlation analysis, regression models, and hypothesis testing.
2. **Variable selection and cleaning** — Only variables relevant to the research questions were retained. These included daily social media usage (hours/day), productivity score (0–10), stress level (1–10), sleep hours per night, coffee intake (cups/day), and burnout days per month. Column names were standardized, and categorical variables such as gender were cleaned to ensure consistent grouping.
3. **Outlier inspection** — Numerical variables were visually examined using boxplots and histograms. Daily social media usage ranged from near zero to approximately 18 hours per day, while burnout days ranged from 0 to about 30 days per month. These values were considered plausible behavioral extremes, and no observations were removed.

The dataset initially contained 30,000 observations and 19 variables. After removing records with missing values, the final cleaned dataset consisted of 21,453 observations and 11 variables.

These preprocessing steps ensure that subsequent analyses, including exploratory data analysis, correlation analysis, t-tests, linear regression, multiple regression, and clustering, are based on clean and consistent data.

4 Exploratory Data Analysis

The purpose of exploratory data analysis (EDA) is to summarize the main characteristics of the dataset and visually inspect distributions, variability, and potential anomalies before applying formal statistical tests.

After preprocessing, the analytical dataset consists of 21,453 observations and 11 variables related to social media usage, productivity, stress, sleep, coffee intake, and burnout.

4.1 Summary Statistics

The dataset exhibits substantial variability across individuals. Daily social media usage ranges from very low values to several hours per day. Productivity scores span the full 0–10 scale, while stress levels also cover the entire measurement range. Lifestyle variables such as sleep hours, coffee intake, and burnout days show heterogeneous patterns, indicating diverse behavioral profiles across users.

4.2 Distributions of Key Variables

Histograms were used to examine the distributions of the main numerical variables.

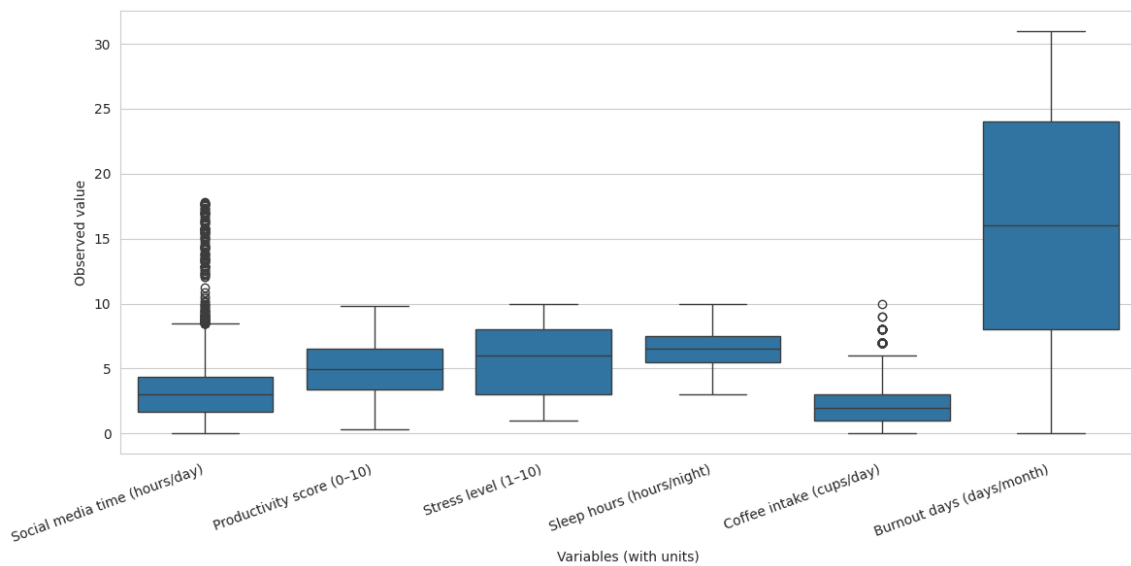


Figure 1: Distributions of key variables

Daily social media usage shows a right-skewed distribution, with most individuals reporting moderate usage and a smaller group exhibiting high usage. Productivity scores are relatively well spread across the scale. Stress levels appear discretized due to their integer-based measurement. Sleep hours are approximately bell-shaped, while coffee intake and burnout days display skewness, reflecting variability in lifestyle behaviors.

4.3 Outlier Inspection

Histograms were used to assess variability and examine the distribution of the main variables.

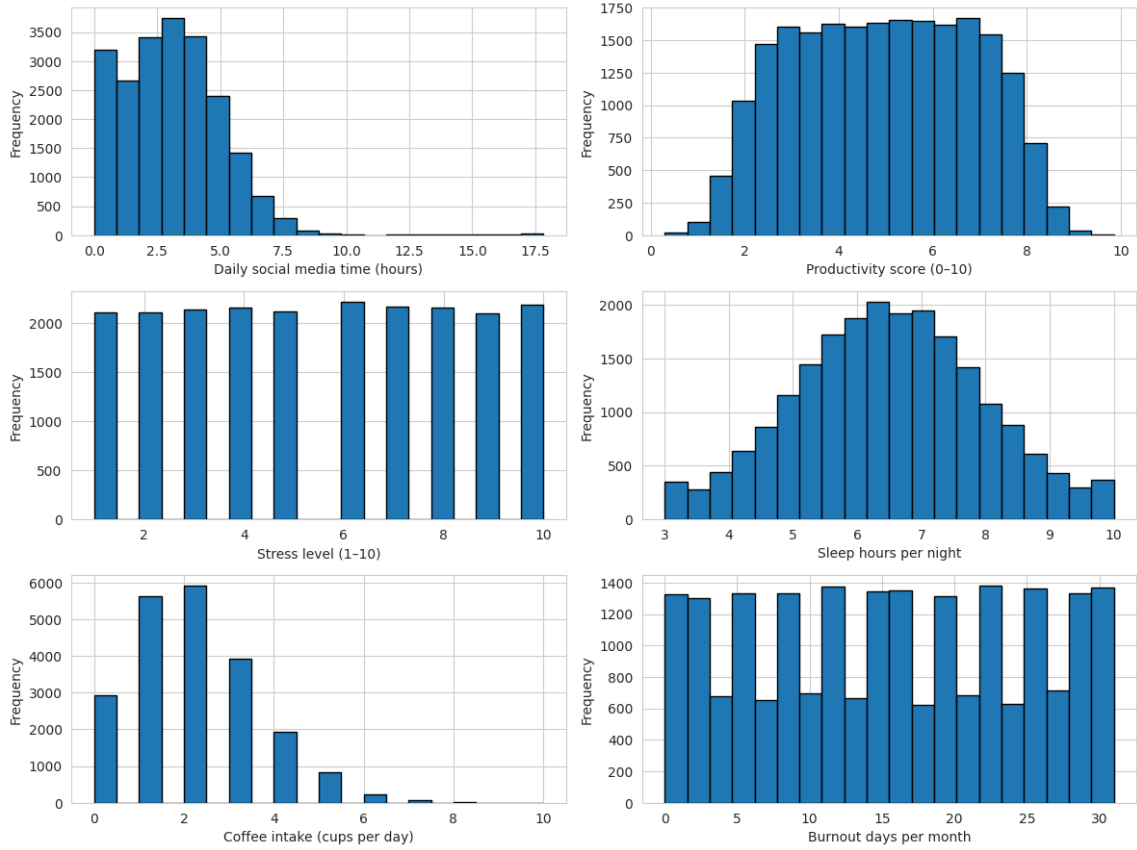


Figure 2: Distributions of key variables (histograms)

The histograms show substantial variability across the main variables. In particular, social media usage and burnout days exhibit right-skewed distributions with extreme values. These observations were retained, as they likely represent realistic behavioral patterns rather than data errors. Their potential influence is examined later using regression-based analyses.

4.4 Correlation Structure

To explore linear relationships between variables, a correlation matrix was computed.

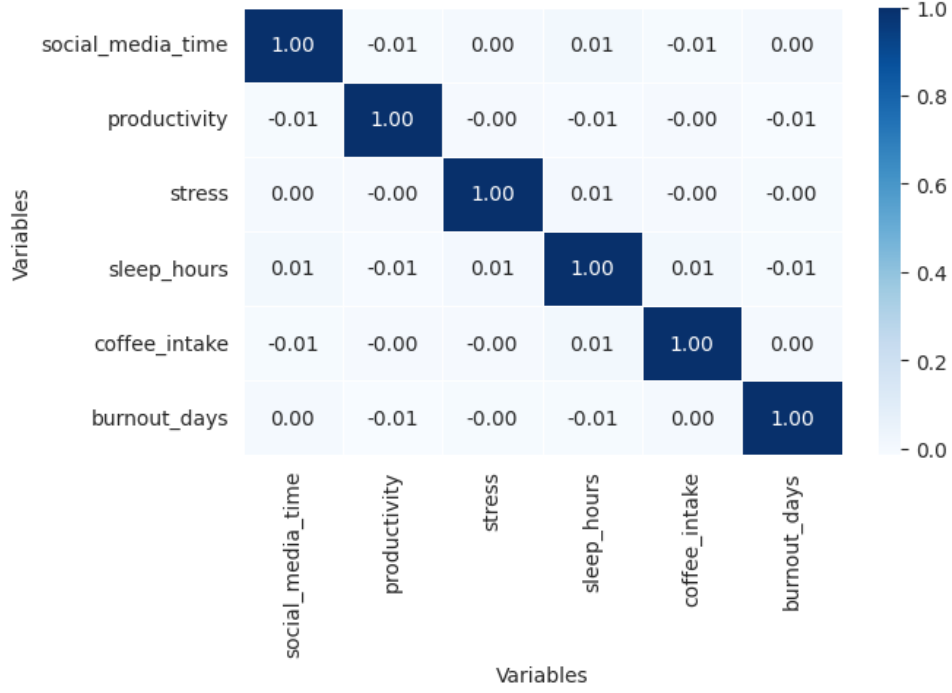


Figure 3: Correlation matrix

The correlation matrix shows that most pairwise correlations are weak. In particular, social media usage exhibits near-zero linear correlation with both productivity and stress, suggesting that any associations may be small or dependent on additional factors. This motivates the use of regression models in subsequent analysis, as prior studies have shown that the relationship between social media usage, productivity, and well-being is often weak or context-dependent [1, 2].

4.5 Key Exploratory Insights

Based on the exploratory analysis, several observations emerge:

- Social media usage varies widely across individuals.
- Productivity and stress span their full measurement ranges.
- Several variables exhibit skewness and extreme values.
- Simple correlations suggest weak direct linear relationships.

These findings align with existing literature indicating that the effects of social media usage on productivity and well-being are generally modest and highly dependent on contextual and individual factors [1, 2]. Prior studies also emphasize that productivity outcomes are influenced by a wide range of psychological and environmental variables beyond simple usage metrics [3].

5 Statistical Modeling and Analysis

After exploratory analysis, several statistical methods were applied to examine the relationships between social media usage, lifestyle factors, productivity, and stress. These methods include simple linear regression, hypothesis testing using t-tests, multivariate regression, and unsupervised learning techniques.

5.1 Simple Linear Regression

Simple linear regression was used to examine bivariate relationships between continuous variables.

First, productivity was regressed on daily social media usage time. The estimated regression coefficient was small and negative ($\beta = -0.0043$) and statistically non-significant ($p = 0.552$). This indicates that social media usage time does not have a meaningful linear association with productivity. The fitted regression line is nearly flat, and the wide dispersion of points around the line confirms the absence of a systematic relationship.

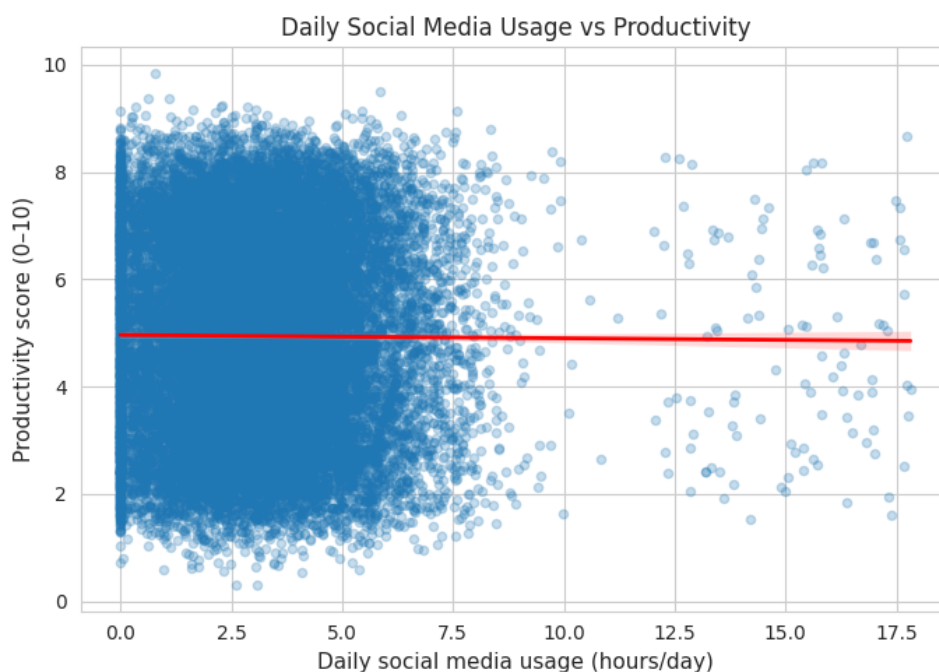


Figure 4: Social media usage time versus productivity with fitted regression line.

A second simple linear regression examined the relationship between job satisfaction and productivity. In contrast to social media usage, the estimated coefficient was positive and statistically significant ($p < 0.001$). The fitted line shows a clear upward trend, indicating that higher job satisfaction is associated with higher productivity levels.

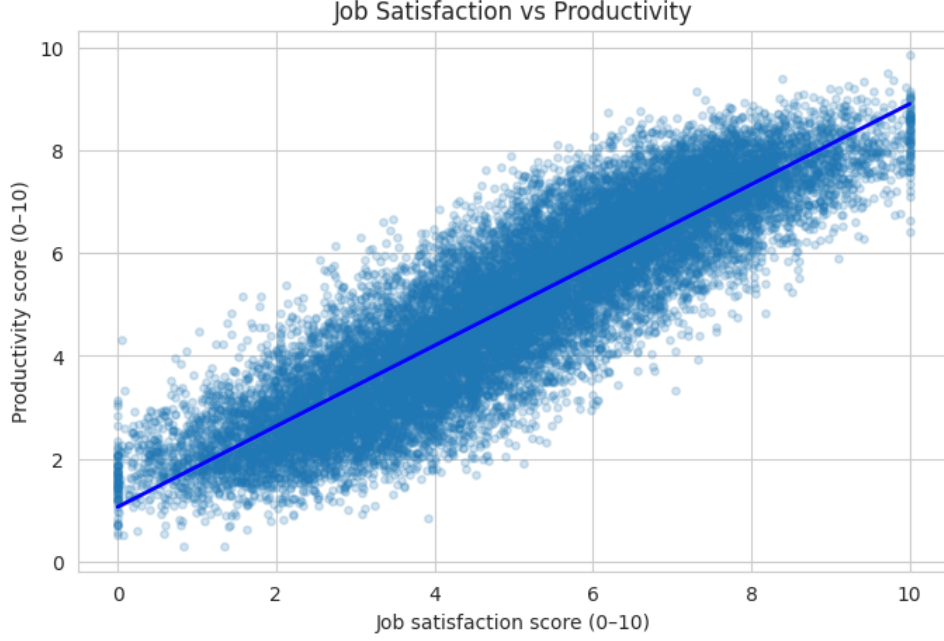


Figure 5: Job satisfaction versus productivity with fitted regression line.

5.2 Two-Sample t-Test

A Welch two-sample t-test was conducted to examine whether mean productivity differs by gender.

Hypotheses:

- H_0 : The mean productivity is the same for male and female participants.
- H_1 : The mean productivity differs between male and female participants.

The male group ($n = 7,717$) had a mean productivity score of 4.944 (SD = 1.882), while the female group ($n = 7,784$) had a mean productivity score of 4.976 (SD = 1.885). The observed mean difference was 0.032 points.

The Welch t-test yielded a test statistic of $t = -1.067$ with a corresponding p -value of 0.286. Since the p -value exceeds the 0.05 significance level, the difference in mean productivity between genders is not statistically significant.

To assess practical significance, Cohen's d was computed and found to be $d = -0.017$, indicating a negligible effect size. This suggests that the observed difference in productivity between males and females is trivial relative to the variability within each group.

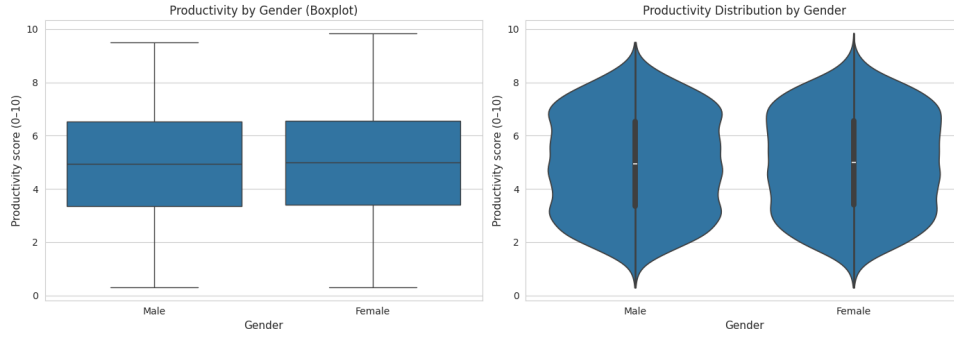


Figure 6: Productivity distributions by gender.

Overall, the null hypothesis is not rejected. These results indicate that gender alone does not meaningfully explain variation in productivity in this dataset.

5.3 Multiple Linear Regression

A multiple linear regression model was fitted to examine whether lifestyle factors jointly predict productivity. The predictors included daily social media usage time, sleep hours, coffee intake, burnout days, and stress level. The overall model fit was weak, with an F -statistic of 1.19 and a non-significant p -value of 0.309 (Table 1), indicating that the model explains virtually none of the variation in productivity.

Predictor	Coefficient	Std. Error	p-value
Social media time	−0.0042	0.0072	0.566
Sleep hours	−0.0155	0.0101	0.125
Coffee intake	−0.0017	0.0104	0.873
Burnout days	−0.0028	0.0016	0.078
Stress level	−0.0021	0.0052	0.685

Table 1: Multiple Linear Regression Results for Lifestyle Factors and Productivity.

$$N = 16,148, R^2 = 0.0004, \text{ Adjusted } R^2 = 0.0001$$

None of the estimated coefficients are statistically significant at the 0.05 level. The extremely low R^2 value confirms that these lifestyle factors jointly explain virtually none of the variation in productivity.

5.4 PCA and Clustering Analysis

Principal Component Analysis was applied to standardized behavioral variables, and the first two principal components were used for visualization. K-means clustering with $k = 3$ was then applied in the reduced PCA space.

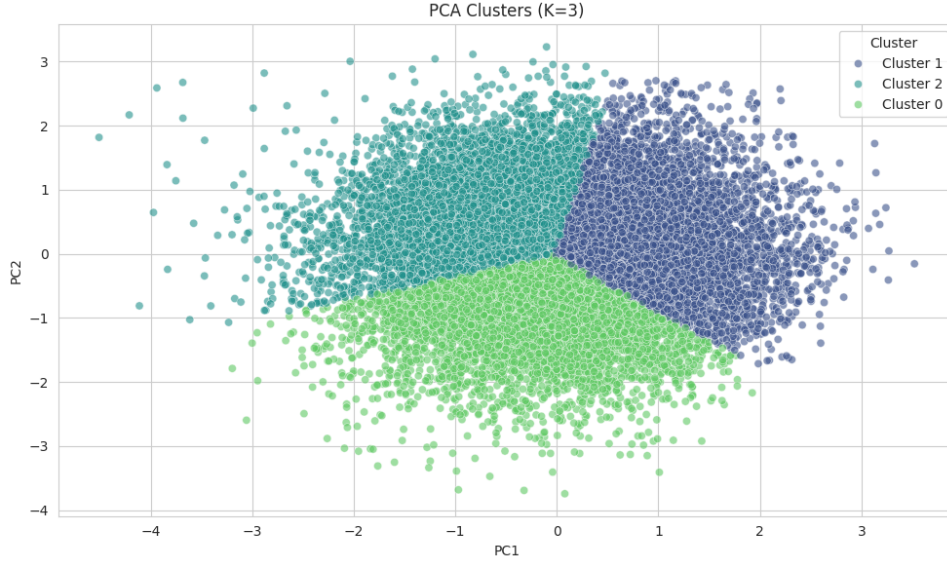


Figure 7: PCA projection with k-means clustering ($k = 3$).

The PCA visualization shows partial separation between clusters, indicating distinct behavioral profiles. These clusters highlight heterogeneity in how individuals combine social media usage, stress, and lifestyle factors, rather than strong linear relationships with productivity.

6 Interpretation and Discussion

The results of this study show that productivity is not well explained by simple lifestyle variables when linear statistical models are used. In particular, social media usage did not show a statistically significant linear relationship with productivity, and the regression models including sleep hours, coffee intake, burnout days, and stress explained very little of the variation in productivity. This indicates that these variables, when considered individually or linearly, are insufficient to capture productivity differences across individuals.

In contrast, job satisfaction displayed a strong and statistically significant positive association with productivity. Individuals with higher job satisfaction consistently showed higher productivity scores. This suggests that productivity is more closely related to psychological and work-related factors than to daily lifestyle habits alone, which helps explain why lifestyle-based regression models performed poorly.

A key limitation of the analysis is the absence of several important factors that were not observed in the dataset. Productivity is likely influenced by job role, workload intensity, work environment, motivation, and individual personality traits. If these unobserved factors are related to lifestyle behaviors, the estimated regression coefficients may be affected by omitted variable bias, reducing their interpretability.

Another challenge is the potential for mutual influence between variables. Stress, burnout, and productivity may affect each other at the same time rather than in one direction. For example, low productivity may increase stress, while high stress may also reduce productivity. This makes causal interpretation difficult when using cross-sectional data. Importantly, the clustering analysis provided insight that was not visible through regression alone. While linear models failed to detect strong relationships, clustering revealed that individuals form distinct behavioral profiles based on combinations of social media use, stress, sleep, burnout, and productivity. This suggests that productivity does not follow a single global pattern, but instead varies across different behavioral groups. Future work could build on these findings by including richer job-related variables, particularly job satisfaction and working conditions. Using interaction terms, non-linear models, or segmenting individuals before modeling may lead to more accurate and meaningful explanations of productivity.

7 Conclusion

This project analyzed how social media usage, lifestyle factors, stress, and job satisfaction relate to productivity using statistical and exploratory techniques. The results showed that social media usage and lifestyle variables do not have strong linear relationships with productivity, while job satisfaction is clearly and positively associated with higher productivity. Gender differences in productivity were not statistically significant, indicating similar productivity patterns across groups. Although regression models explained very little variation, the clustering analysis revealed distinct behavioral profiles, highlighting that productivity differs across groups rather than following a single linear trend. Overall, the findings show that productivity is shaped by complex behavioral and work-related factors. Simple linear models are useful for testing basic relationships, but exploratory methods such as clustering are necessary to better understand heterogeneous behavioral patterns in productivity data.

8 References

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9 Appendix A: Code Implementation

All data preprocessing, exploratory analysis, regression modeling, and clustering were implemented using Python. The complete code for this project is available at:

<https://colab.research.google.com/drive/1Zd2bvJwkuKwDn1Y1GC3lN9Cn4o6j5ru5?usp=sharing>

10 Appendix B: Use of AI Assistance

ChatGPT (<https://chatgpt.com/c/693d3093-afc8-8326-90ea-cb9ade902750>): moral support, formatting help, an AI-generated cover image, and zero data analysis.